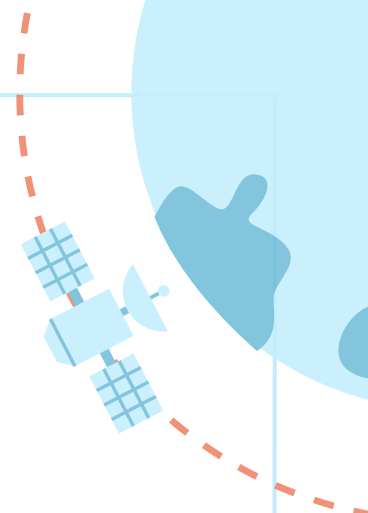


# Impact of DEM Bit Depth on Slope Analysis

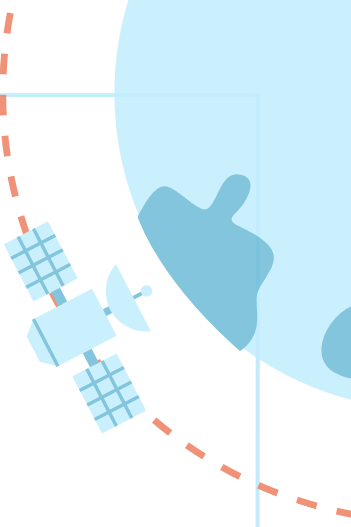
*Case Studies: Kerala 🇮🇳 & Abu Dhabi 🇦🇪*



## Why This Study? :

- DEM bit depth (8-bit vs 32-bit) changes how slopes are calculated.
- But... does it really impact accuracy?
- To find out, I tested Kerala 🇮🇳 and Abu Dhabi 🇦🇪.
- I compared results with statistics + maps.

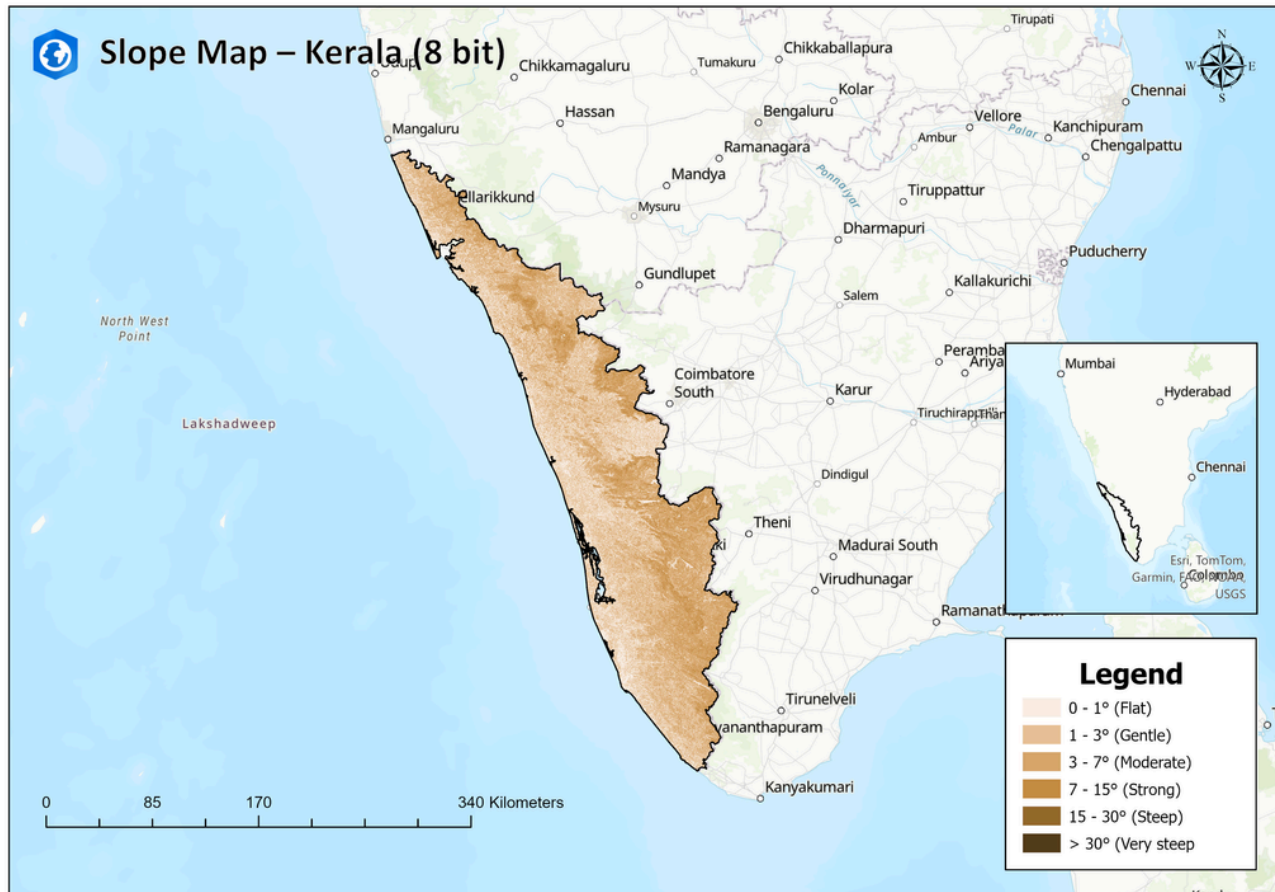




## How I Did It? :

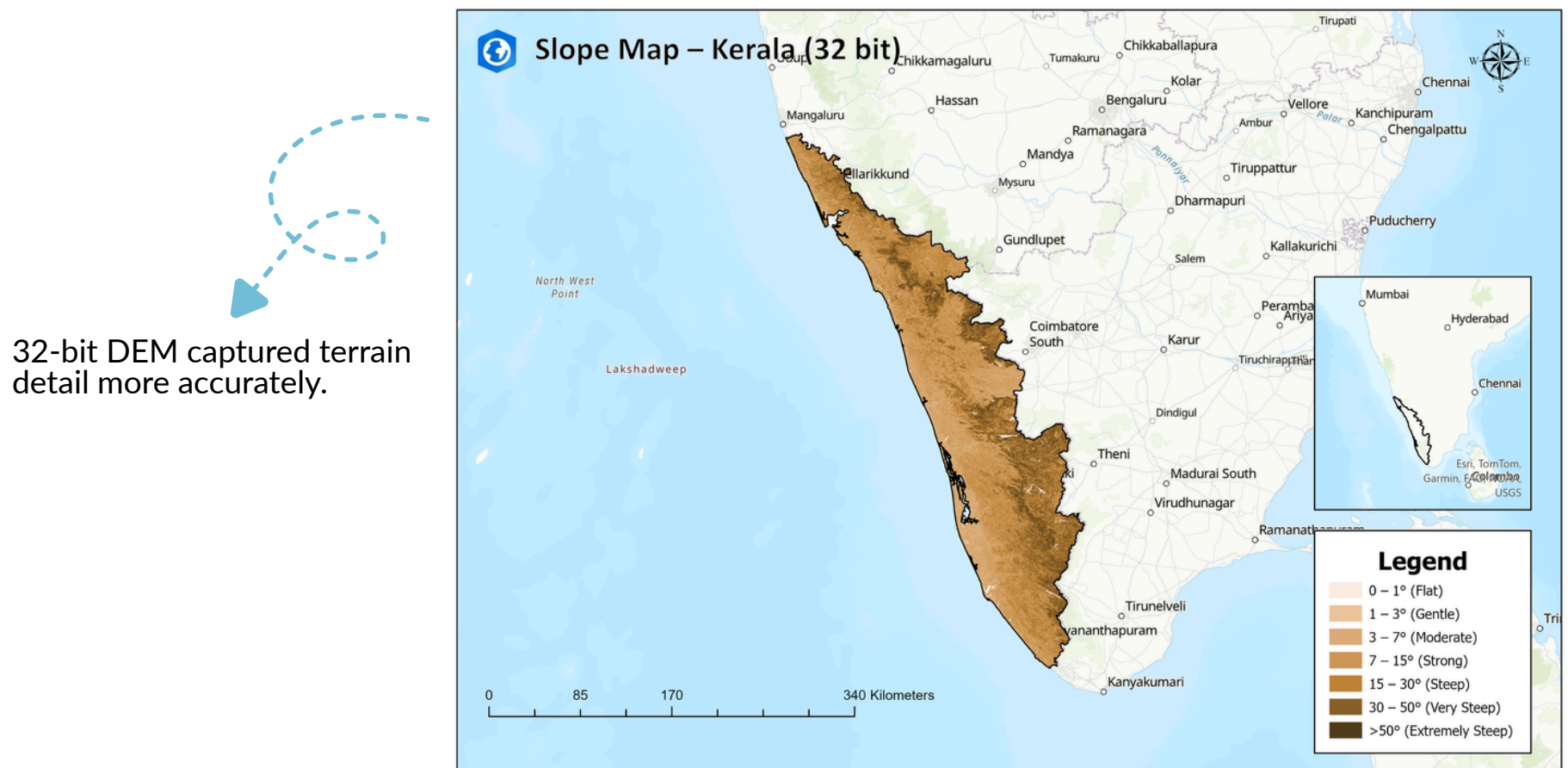
- First collected DEM datasets in different bit depths (8-bit & 32-bit).
- Then processed them in ArcGIS Pro (slope analysis, classification, statistics).
- I focused on two case studies:
  - i. Kerala, India 
  - ii. Abu Dhabi, UAE 
- I compared slope outputs using maps + RMSE, MAE,  $R^2$  metrics.

## Case Study: Kerala, India



8-bit DEM resulted in loss of slope variation in hilly regions.

- The Western Ghats create steep slope zones.
- Visual differences highlight how bit depth impacts slope outputs.

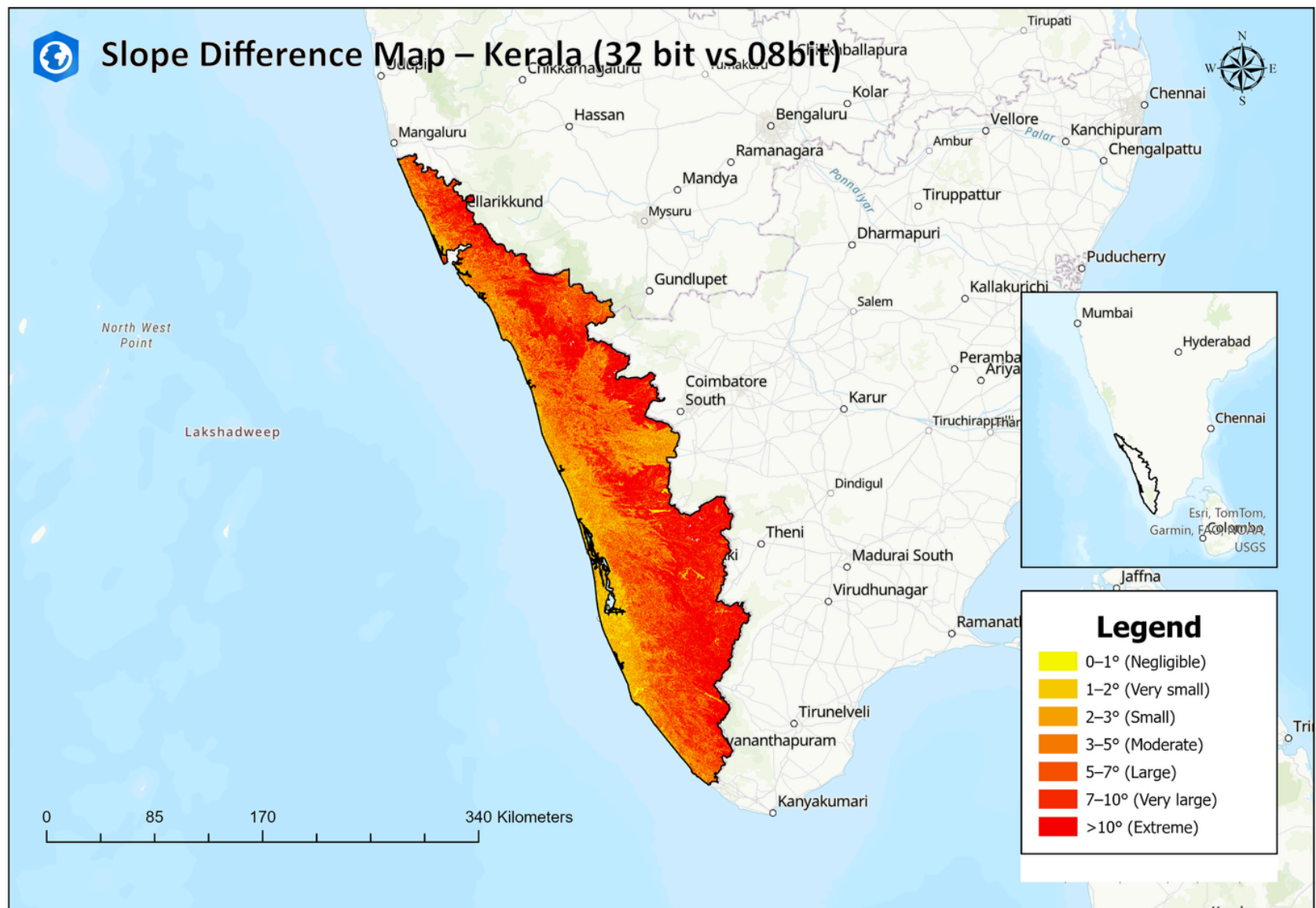


32-bit DEM captured terrain detail more accurately.



# Case Study: Kerala, India

Negligible differences in flat coastal plains

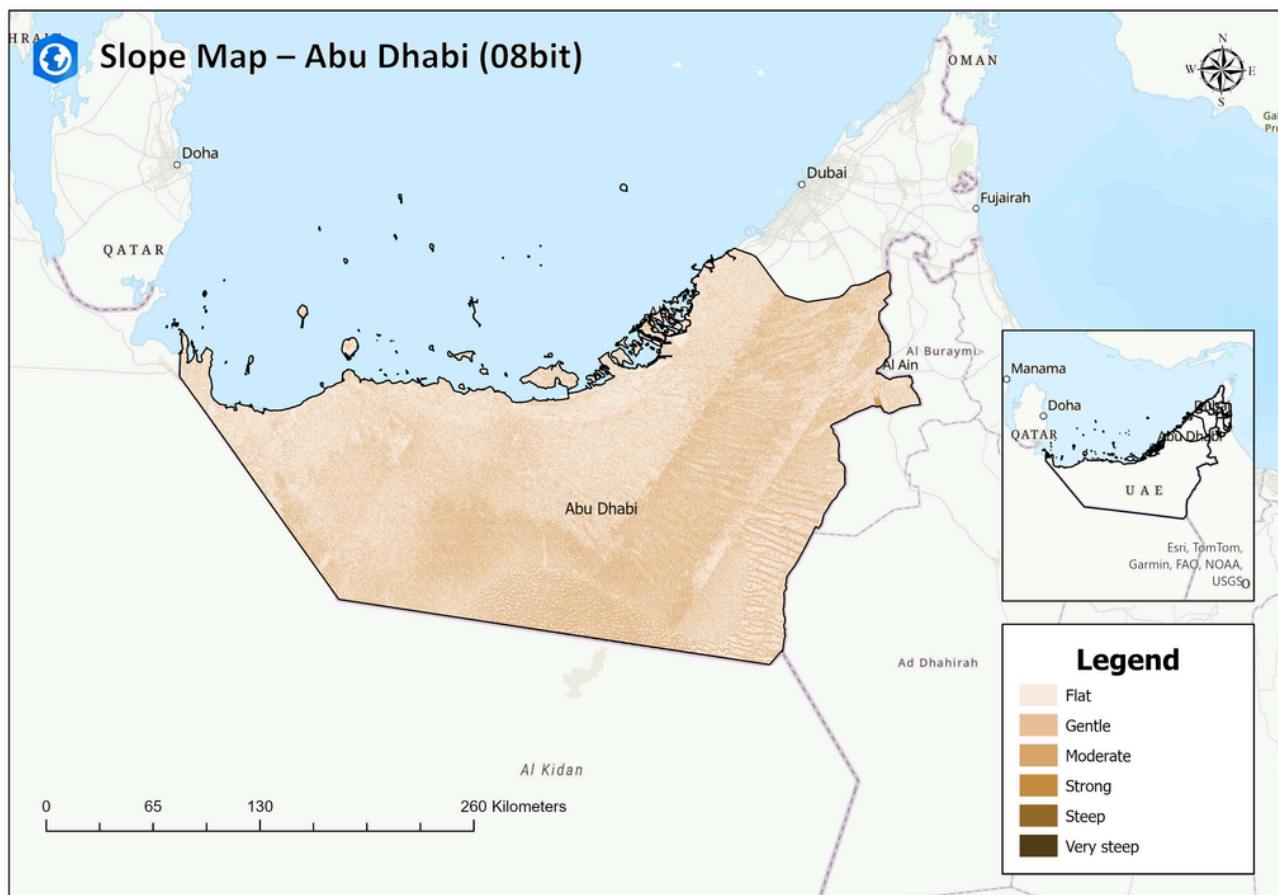


Large discrepancies in Western Ghats — 8-bit fails to capture steep slopes

- Differences are minimal in flat terrain, significant in hilly regions.
- Highlights loss of accuracy when using lower bit-depth DEMs.



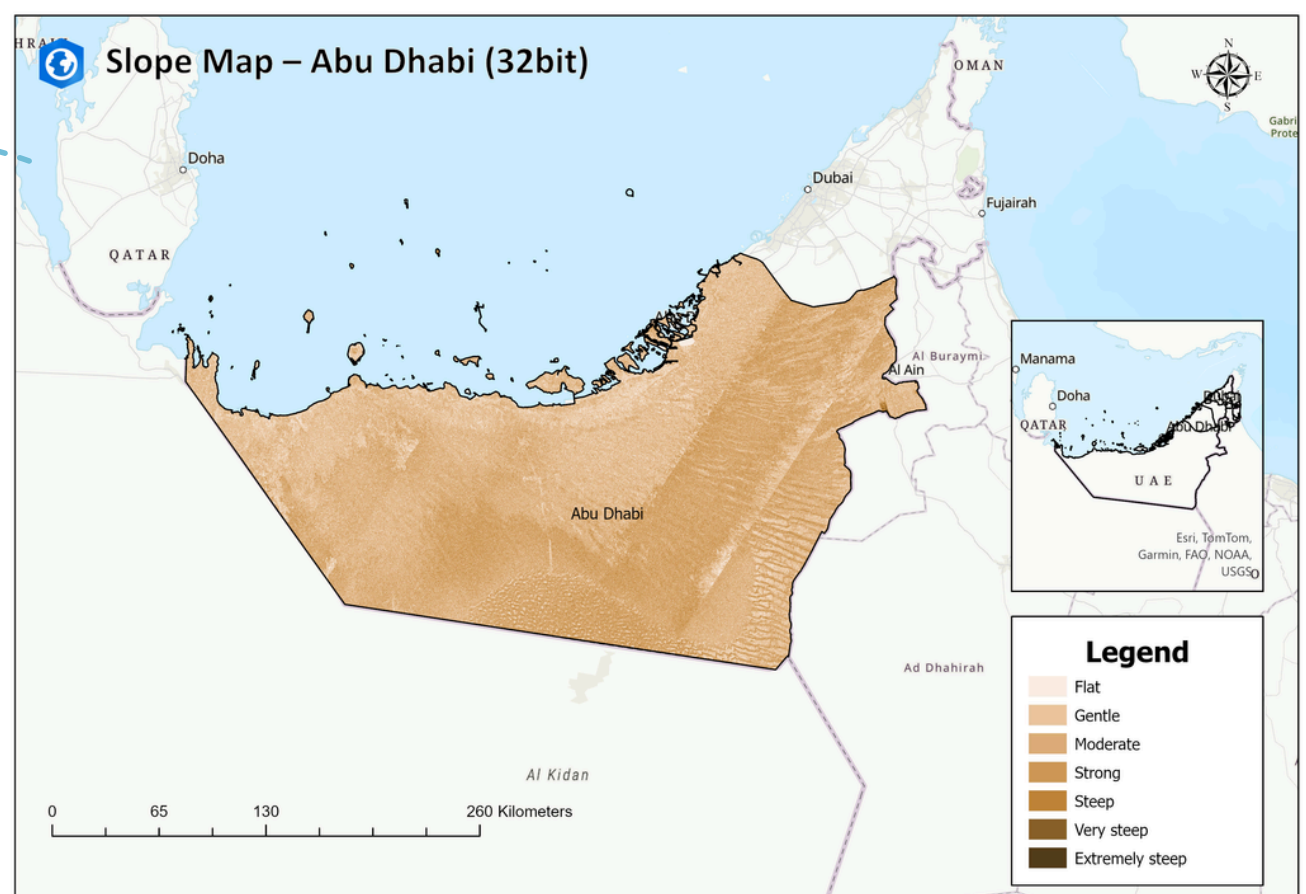
## Case Study: Abu Dhabi, UAE



8-bit DEM flattens terrain in desert & coastal areas.

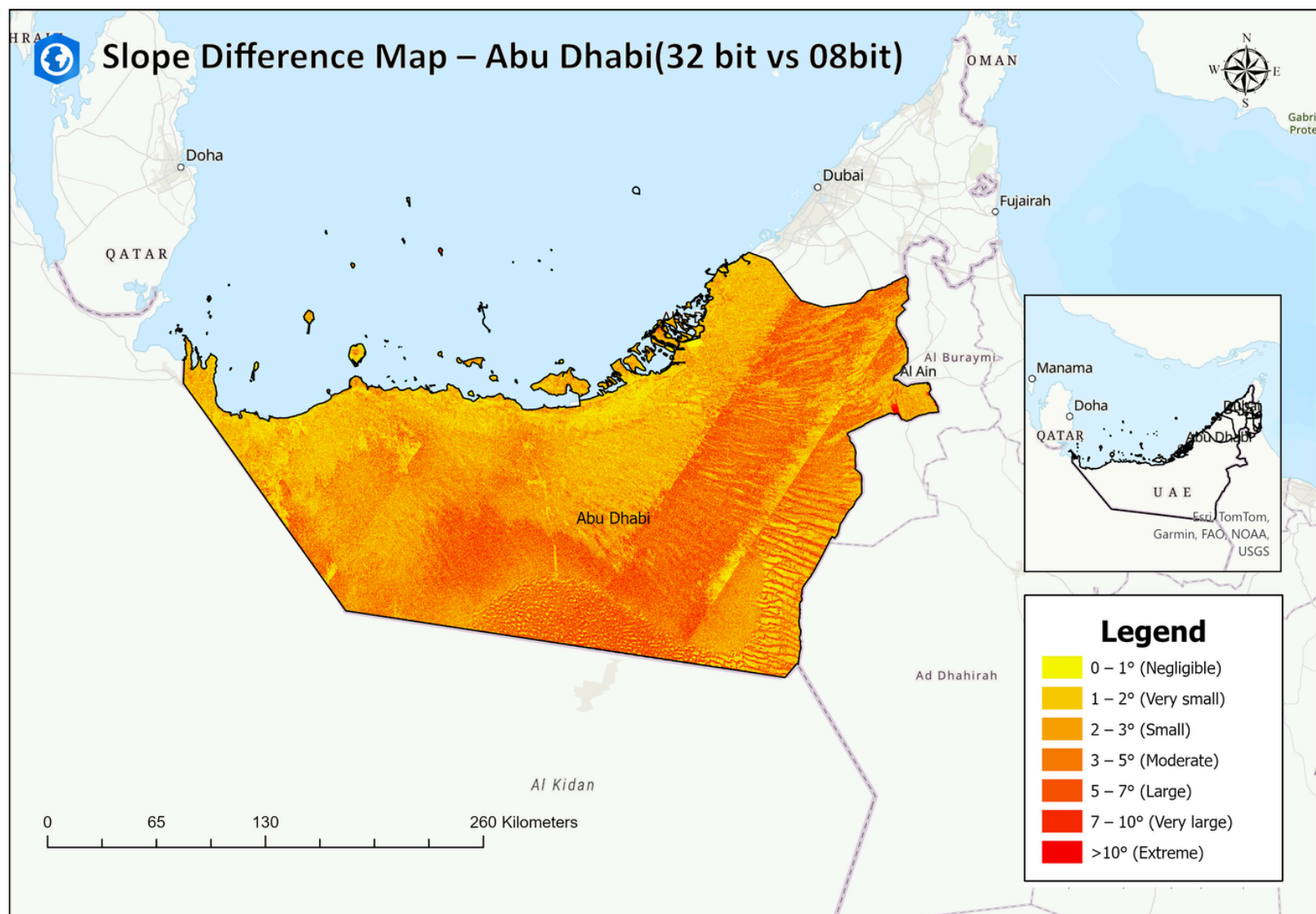
- Terrain in Abu Dhabi is generally flat, but subtle slope variations matter.
- Higher bit depth DEM captures micro-variations critical for engineering & hydrology.

32-bit DEM preserves subtle variations in dunes & escarpments.



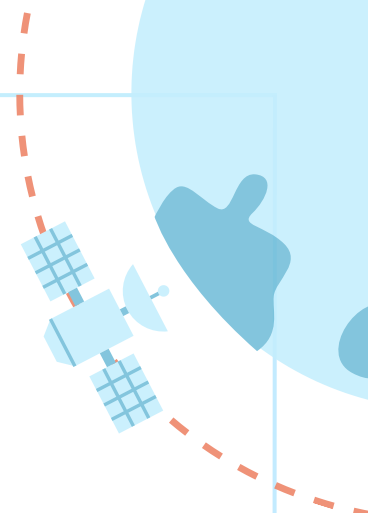
## Case Study: Abu Dhabi, UAE

Negligible differences in flat desert regions.



Subtle but important differences in dune & escarpment zones.

- In flat regions, bit depth differences are minimal.
- Localized terrain features (sand dunes, escarpments) show variation between 8-bit and 32-bit.
- Even small discrepancies can influence hydrological or engineering modeling in arid regions.



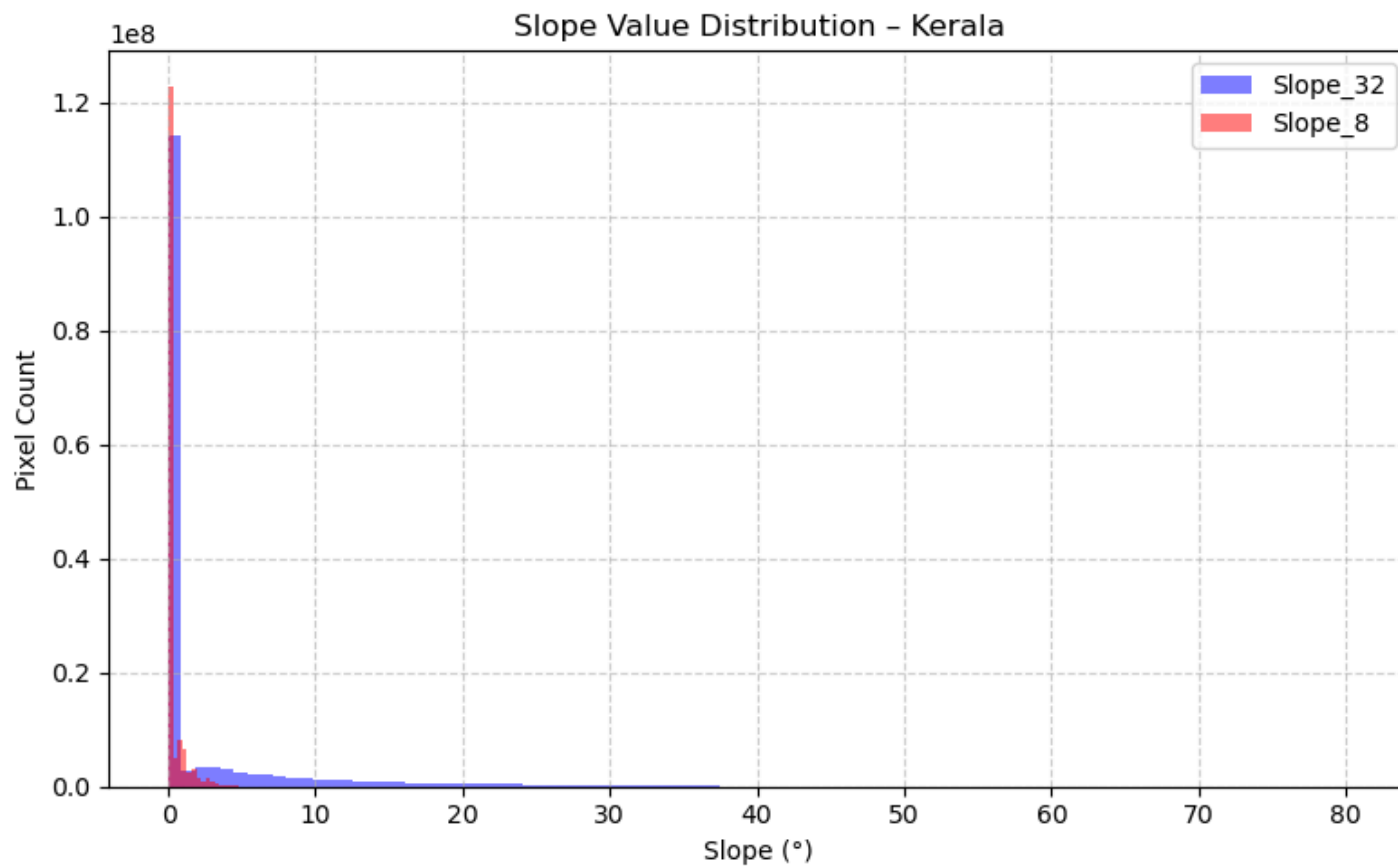
## Error Metrics (Slope 32-bit vs 8-bit)

| Metric         | Kerala | Abu Dhabi |
|----------------|--------|-----------|
| RMSE           | 6.17   | 3.05      |
| MAE            | 2.38   | 1.56      |
| R <sup>2</sup> | 0.93   | 0.90      |

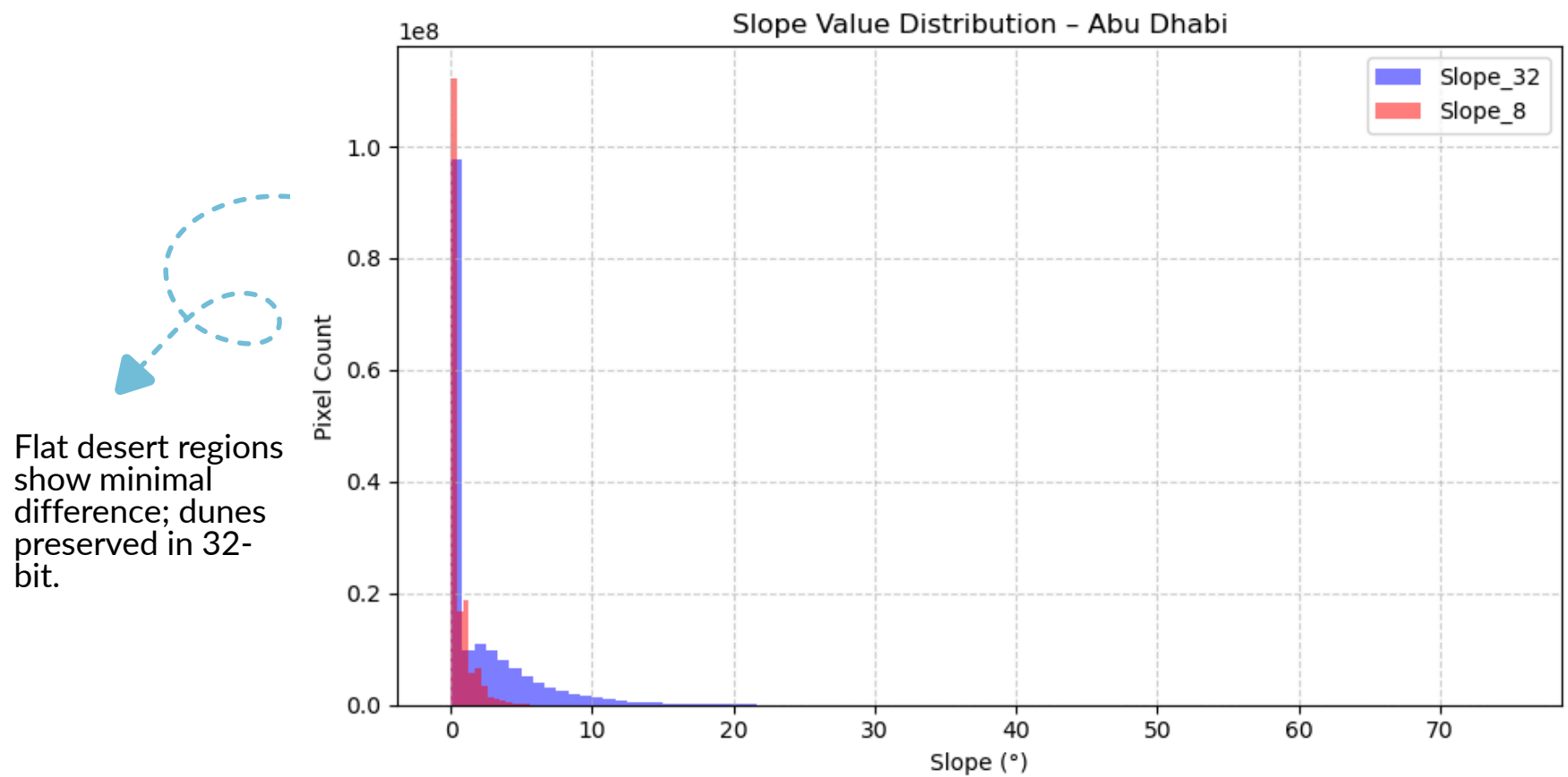




# Slope Distribution Comparison



Western Ghats  
create steep  
variations; 8-bit  
loses detail.

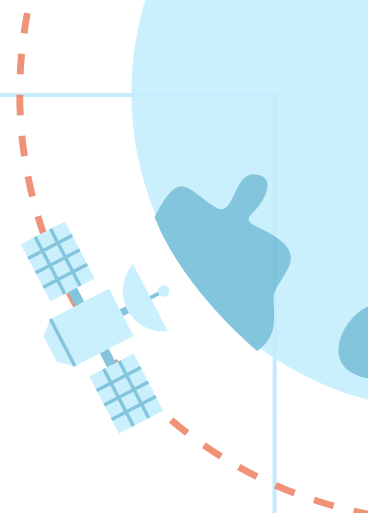


Flat desert regions  
show minimal  
difference; dunes  
preserved in 32-  
bit.



## Key Insights from DEM Bit Depth Comparison

- 🌍 Flat Terrain → "Negligible differences"
- 🏔️ Mountains/Western Ghats → "Major slope variation loss in 8-bit"
- 🌵 Deserts/Dunes (Abu Dhabi) → "Subtle but important micro-variations"
- Minimal accuracy loss in flat regions.
- Steep terrain shows large slope discrepancies.
- 32-bit DEM preserves detail for hydrology, engineering & risk modeling.
- Small errors in DEM resolution can scale up in decision-making.



## Conclusion

**“DEM bit depth significantly affects slope analysis accuracy.”**

- Flat and coastal areas show minimal differences across bit depths.
- Steep terrains (e.g., Western Ghats) highlight major discrepancies in 8-bit outputs.
- Higher bit depth (32-bit) ensures better precision for engineering, hydrology, and environmental modeling.

**Choosing the appropriate DEM bit depth is critical for reliable spatial decision-making.**

